

# tmux documentation

## Tmux colors customizations

How can we change the bottom green panel from green color to some other in tmux? Is it possible to be parametrized or we have to modify source code?

### Answer

To change the color of that green bottom bar (called the **status line**), you don't need to modify any source code. It's completely configurable through tmux's own options. You can set the color directly from the command line, or make it permanent in your `~/.tmux.conf` file.

#### The Quick Way (temporary change)

1. Press your prefix key (usually `Ctrl+b`), then type `:` to enter command mode.
2. Type the following command and press Enter:

```
set -g status-style bg=blue,fg=white
```

Replace `blue` and `white` with whatever colors you prefer. For example, to replicate the classic black background:

```
set -g status-style bg=black,fg=white
```

#### The Permanent Way (through your config file)

Add this line to your `~/.tmux.conf` file:

```
# Set status bar background and foreground colors
set -g status-style 'bg=black,fg=white'
```

After saving the file, reload the configuration with:

```
tmux source-file ~/.tmux.conf
```

Or, from within tmux, use `Prefix + :` and run:

```
source-file ~/.tmux.conf
```

#### Available Color Options

You can use:

- **Basic color names:** `black`, `red`, `green`, `yellow`, `blue`, `magenta`, `cyan`, `white`
- **Extended 256-color palette:** `colour0` to `colour255` (e.g., `colour240` is a nice dark gray)
- **Hexadecimal RGB values:** `#ff0000` (requires terminal support for true color)
- `default`: makes the status bar transparent (inherits the terminal's background)

Example:

```
set -g status-style 'bg=colour240,fg=colour255'
```

## 💡 Making It Dynamic (Parameterization)

Yes, you can dynamically change the status bar color based on conditions. Tmux supports `%i f` statements and format variables. For instance, this snippet changes the bar to red when running on a host named “myhost”, green on “myotherhost”, or blue otherwise:

```
%if "#{==:#{host},myhost}"
  set -g status-style bg=red
%elif "#{==:#{host},myotherhost}"
  set -g status-style bg=green
%else
  set -g status-style bg=blue
%endif
```

You can even use shell commands or environment variables to set the color, for example:

```
run "tmux set -g status-style bg=$MY_TMUX_COLOR"
```

## 👁 Hiding the Status Bar Altogether

If you’d rather not see the bar at all, add this to your config:

```
set -g status off
```

## 📖 Further Reading

- [tmux man page \(status-style\)](#) – official documentation.
- [tmux bottom status bar color change \(Unix & Linux Stack Exchange\)](#) – community discussion with multiple methods.
- [tmux の status line の色を変更 \(Qiita\)](#) – a clear, step-by-step guide.

So, to answer your question directly: **No source modification is required.** Everything you need is already built into tmux’s configuration system.

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## Implementation: Hostname-Derived Status-Bar Color

The vasic-digital tmux project implements an algorithm that maps each host's hostname to a deterministic colour from a curated 27-colour palette, making visually distinguishable tmux sessions across different connected machines.

### Algorithm

File: `scripts/hostname_color.sh`

1. **DJB2 hash** of the hostname string (31-bit, positive)
2. **Palette index** = hash modulo palette size (27)
3. **Output** = colourNNN (xterm 256-colour name)

The palette selects visually distinct colours that work well as `tmux status-bg` backgrounds with white foreground text:

```
colour1    colour3    colour4    colour5    colour6
colour9    colour11   colour12   colour13   colour14
colour52   colour88    colour130  colour166  colour172
colour178  colour190  colour196  colour198  colour199
colour200  colour202  colour208  colour214  colour220
colour226  colour240
```

## Integration

The `scripts/tmx` wrapper invokes `hostname_color.sh` after the `tmux` server starts and applies the colour via:

```
tmux set -g status-style bg=<colour>
```

The colour is set both on session creation (`new-session`, `start-server`) and on re-attach, so it always reflects the current host.

## Verification

| Test                             | Scope                                                                                             | Evidence                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 10_hostname_color_algorithm.sh   | 5 invariants:<br>deterministic, valid<br>format, palette<br>membership, spread,<br>empty-fallback | Captured hash/<br>index/colour output                                            |
| 11_hostname_color_integration.sh | Wrapper applies correct<br>colour to <code>tmux</code> server                                     | <code>show -g<br/>status-style</code><br>readback matched to<br>algorithm output |

## Usage

```
# Manually query the colour for any hostname
bash scripts/hostname_color.sh                # colour52
      (this host)
bash scripts/hostname_color.sh myserver.example.com # colour166
```

## Anti-bluff covenant (Constitution 1)

- Every PASS in both tests carries positive runtime evidence:
  - The algorithm test prints hash, index, and colour for known inputs.
  - The integration test reads `status-style` back from the running `tmux` server and compares it to the algorithm's expected output.
- The colour computation is deterministic -- same hostname always produces the same colour. No randomness. No state.
- FAIL-bluff protection: all scripts use `set -uo pipefail` and guard every variable reference.